ITM SKILLS UNIVERSITY

School of Future Tech

Case Study Report

on

Voting Eligibility Checker

by

Pari Gothi
150096725097

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1. Introduction to the Case Study

Age-based eligibility verification is a fundamental requirement in democratic systems. In elections, only citizens who meet the minimum age requirement are permitted to vote. This case study focuses on designing and implementing a Voting Eligibility Checker program in C++ that determines whether a person is eligible to vote based on their age.

The case study demonstrates how conditional logic, while loops, and counter variables can be combined in C++ to build a practical, interactive console application. It highlights the importance of input validation, control flow, and accumulating results across multiple iterations.

2. Problem Statement / Case Background (Abstract)

Background

In many real-world scenarios, software systems need to check eligibility conditions for large numbers of individuals. A simple, reliable, and efficient program is needed that can accept multiple age inputs, validate each one against a defined threshold, and report the total number of eligible individuals.

Abstract

This case study presents the design and implementation of a Voting Eligibility Checker using C++. The program accepts the number of persons as input, iterates through each person's age using a while loop, applies conditional logic (if-else) to determine eligibility based on the age threshold of 18, and maintains a counter variable (eligibleVoter) to accumulate the total number of eligible voters. The final count is displayed at the end of execution. The implementation demonstrates mastery of conditional logic, loops, and variable management in C++.

3. Case Study Design

The Voting Eligibility Checker is designed as a simple sequential program with the following stages:

- **Input Phase:** The user is asked to enter the total number of persons to be checked.
- **Iteration Phase:** A while loop runs as long as i < n, prompting the user to enter each person's age.
- **Validation Phase:** An if-else block checks whether the entered age is greater than or equal to 18.
- **Counting Phase:** A counter variable (eligibleVoter) is incremented each time an eligible voter is found.
- **Output Phase:** After the loop ends, the program displays the total number of eligible voters.

project.cpp

project.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main(){
  int age, n;
  int eligibleVoter = 0;
  int i = 0;
  cout <<"Enter the number of people: ";
  cin >> n;
  while (i < n)
  {
    cout <<"Enter your age: ";
    cin >>age;

    if (age >= 18)
    {
      cout <<"You're eligible to vote\n";
      eligibleVoter++;

    }
    else
    {
      cout <<"You're not eligible to vote\n";
    }
    i++;
  }

  cout <<"Total eligible voter = " <<eligibleVoter;



  return 0;
}
```

4. Methods & Algorithms / Technology Applied

Key Concepts Used:

- **Variables and Data Types:** Integer variables used for age, count (n), loop counter (i), and eligibleVoter (initialized to 0).
- **Input/Output:** cin and cout used for console-based interaction.
- **While Loop:** Runs as long as i < n. The counter i is manually incremented at the end of each iteration using i++.
- **Conditional Logic (if-else):** Checks if age >= 18 to determine eligibility.
- **Counter Variable:** eligibleVoter incremented inside the if block to count eligible voters.

Technology Stack:

- Programming Language: C++
- Standard Library: <iostream>
- Compiler: g++ / any standard C++ compiler
- Environment: Any C++ IDE (Code::Blocks, VS Code, Dev-C++, etc.)

5. Implementation Details

The program is implemented as a single C++ source file. Below is the structure of the implementation:

Variable Declarations:

Four integer variables are declared: n (number of people), age (current person's age), i (loop counter initialized to 0), and eligibleVoter (initialized to 0).

User Input for n:

The program prompts the user to enter the number of people and stores the value in n using cin.

While Loop:

A while loop runs as long as i < n. In each iteration, the user is prompted to enter an age. At the end of each iteration, i++ increments the counter to move to the next person.

Eligibility Check:

Inside the loop, an if-else checks if the age is >= 18. If true, it prints 'You are eligible to vote' and increments eligibleVoter. Otherwise, it prints 'You are not eligible to vote'.

Final Output:

After the loop ends, the program prints the total eligible voter count stored in eligibleVoter.

```
Enter the number of people: 3
Enter your age: 17
You're not eligible to vote
Enter your age: 23
You're eligible to vote
Enter your age: 97
You're eligible to vote
Total eligible voter = 2
```

6. Results and Conclusion

Findings:

- The program correctly identifies eligible and ineligible voters based on the age condition.
- The counter variable eligibleVoter accurately tracks the total number of eligible voters across all iterations.
- The use of a while loop with a manually incremented counter i ensures all n persons are checked.
- Correct placement of eligibleVoter++ inside the if block is critical for accurate results.

Conclusion:

This case study demonstrates a complete implementation of a Voting Eligibility Checker in C++. It successfully integrates user input, while loop-based iteration, conditional logic, and counter-based accumulation into a working console program. The exercise builds a strong foundation in conditional logic mastery, which is the stated learning outcome of Case Study 16.

7. References

- cplusplus.com — Reference documentation for C++ standard library and syntax.
- GeeksforGeeks — Tutorials on C++ loops, conditionals, and variables.